

Benjamin Andrew

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Education

- University of Manchester**, PhD in Computer Science Sep 2024 – Present
- Project titled *Developing Reliable Software for Autonomous Robots*.
 - Supervised by [Dr. Louise Dennis](#), [Dr. Marie Farrell](#), and [Prof. Michael Fisher](#).
- University of Cambridge**, BA (Hons) in Computer Science Oct 2019 – Jun 2022
- Grade II.1 (70%).
 - Dissertation titled *Delay-Tolerant Link-State Routing*. Implemented and empirically evaluated a failure-tolerant routing protocol in C to run on real Unix-based routers, comparing with traditional protocols. Supervised by [Dr. Jackson Woodruff](#).

Research Interests

My interests lie in the intersection of autonomous robotics and formal methods. Specifically, I am passionate about verifying mission-critical systems under uncertainty and failure.

Professional Activities

- Part of the local organising committee of [The 19th International Conference on Integrated Formal Methods 2024](#).
- Member of the [Autonomy and Verification Network](#).

Teaching Experience

- Graduate Teaching Assistant**, University of Manchester
- COMP11120 *Mathematical Techniques for Computer Science* Sep 2024 – Present
 - COMP21111 *Logic and Modelling* Sep 2024 – Dec 2024
- Supervisor** (TA equiv.), University of Cambridge
- CST II *Hoare Logic and Model Checking* Apr 2023 – Jun 2024
 - CST IB *Logic and Proof* Jan 2023 – Mar 2024
 - CST IB *Semantics of Programming Languages* Oct 2023 – Dec 2023
 - CST IA *Introduction to Graphics* Oct 2022 – Dec 2022

Industry Experience

- Software Engineer**, Tarides – Paris, France Sep 2022 – Jun 2024
- Developed open-source CI (Continuous Integration) software for the OCaml language's ecosystem, involving systems programming to distribute dependency solving, compilation, and testing to many operating systems and architectures.
 - Responsible for the CI of the Opam Repository, the central package universe for OCaml with tens of thousands of packages.
- Embedded Systems Intern**, Tarides – Paris, France Jul 2021 – Sep 2021
- Research & development project to cross-compile the OCaml bytecode runtime to an ARM-based microcontroller using a real-time operating system as a base layer.
- Computer Graphics Research Intern**, University of Cambridge Jun 2020 – Sep 2020
- Developed and implemented a visual model of ageing vision in virtual reality with Unity, as part of the Graphics and Displays Research Group.

Volunteering

Run Leader, Runwild – University of Manchester

Dec 2024 – Present

- Lead bi-weekly group runs for a student running group.

English Teacher, Connections English Institute – Ibagué, Colombia

Jul 2024 – Aug 2024

- Taught adult beginners in morning and night classes.